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The development of a function in R language to process diet analysis

Introduction

Diet studies are a crucial point in ecology; it is through diet studies that we can understand the trophic position of an organism and how it relates to the environment (Sabo & Gerber, 2021). Usually, in diet studies, researchers use several types of mechanisms to understand an organism's diet.

When we work on data analysis, we can get information such as how many times a food item was found in the stomachs of a population, as well as the volume of that food item in this population. However, only quantifying how many times an item is found in the stomachs does not provide us with a full explanation. The food item could be small, so if we just analyze how many times that food item occurs, we will reach a result that presents that food item as the most important for that population, as it happens in all stomachs, but this parameter could not represent the real diet of that population. On the other hand, if we analyze the volume of that food item in the stomach of a population by only using the volume, it does not provide a good explanation, as it could still be a rare item. The ideal method is to combine both occurrence and volume to calculate a much better explanation and help us better understand the real data of that sample population.

The Importance Alimentary index (IAi; Kawakami & Vazzoler, 1980) is a tool widely utilized that explains the importance of a food resource on an organism's diet. It takes into consideration the percentage of volume of a food item (V%) and the occurrence of it within the fish population (FO%). For example, a food item that has a high IAi is a particularly important alimentary food item, as the index is a correlation between how much a given food item occupies the stomach and how often it happens in the stomach of the population.

The Importance Alimentary index assumes that there is a relation between the products of the frequency of occurrence percentage and the volume percentage of each item, that can be used to estimate their diet importance by adding them on a cartesian plane, where X is FO% and Y is V%, we can have an area that will be the IAi Σi is summatory of FO% times VO% of all stomachs of the population (Kawakami & Vazzoler, 1980), it can be observed on the following expression:

$$IA_i = (F_i \times V_i) / \Sigma_{i=1}^{n}(F_i \times V_i)$$

where:

- IA_i = food index of item *i*
- F_i = frequency of occurrence (%) of item *i*
- V_i = volume (%) of item *i*
- n = total number of food items analyzed

So, the Alimentary index allows us to correlate the occurrence and the volume and understand better the importance of each food item in a population.

Cleaning Protocols

When data is sampled from fish stomachs, it is usually written in paper notes and transferred to spreadsheets on a computer; the process of transferring data from field to lab work often adds challenges to the dataset, such as unorganized data and repetitive generation of information in spreadsheets and in R due to a lack of data-handling skills, which could lead to disorganization and potential mistakes or typos. Also, sometimes it is not even inputted by the same scientist, which could lead to a lack of standardization of information. In these cases, that spreadsheet needs to go through a cleaning process, which, oftentimes, is not documented, leading to loss of data.

The data cleaning process acts as a 'saviour' that turns the mass of data into something usable. It is a very common issue to have someone spending a lot of time on manual data cleaning and running into the risk of modifying the data and losing information in the process. A Data-cleaning script is a very important tool to retain the original raw data and return the cleaned version while preserving the data history. For modern reproducible science, an automated protocol tool is crucial (Diederich et al., 2022). It is also important for this automated protocol to be reproducible, so an automated protocol could be beneficial to scientists.

Nowadays, there is no automated protocol to calculate the IAi. To avoid the above-mentioned problems, a tool that could automatically calculate the IAi would bring an advantage, speeding the process and allowing the scientist to go to the next step faster than usual, while making the whole data analysis clear and reproducible.

The philosophy of the book R Packages, written by Wickham & Bryan and last released in 2025, states that "Anything that can be automated should be automated." It continues, addressing the need for functions that do more in less time, as it will perform tasks through a trackable protocol, reinforcing good science practices.

Results

With that in mind, seven functions were developed in R, as a first step into automating the process of data analysis:

- validate_diet_data
- clean_diet_data
- run_all_observed_scnarios
- fo_summary
- volume_summary
- iai_summary
- diet_indices_summary

In every function, there is a step for “data frame” verification to prevent consequential errors.

This project was made using the following libraries:

- tidyverse
- janitor

The workflow starts with a validation function that screens the raw database for inconsistencies and data quality issues and returns a report, followed by a cleaning process that attempts to solve all common problems, such as non-standardization of character encoding. This is specifically important due to different character encoding that comes from multiple computer operating systems; one example is the difference between Windows and IOS environments, where one character has multiple ways of being encoded, raising problems of interpretation at the processor level, where it will read this character in two different ways. This is why I am choosing UTF-8 as the text encoding for its simplicity and compatibility with multiple environments. At this point, the validation step will run again to assess if the problems were reduced or fixed. If the validation step is cleared, then we continue to the analysis step.

For the data analysis function, it follows the sequence of:

- It identifies every observed combination of the selected metadata variables.

- It filters the dataset for one combination at a time.

- It sends that filtered subset to fo_summary().

- It sends the same subset to volume_summary().

- It sends the same subset to iai_summary().

- It sends the same subset to diet_indices_summary().

- It stores all outputs under the corresponding scenario name.

To use the script, simply clone the *rod_BIOI835AQ* repository from *Soares Quantitative Ecology Lab*, install (if not already installed) *tidyverse* and *janitor*, refer to the functions script with *“source(“01_script/functions.R”)*, load the raw dataset, run the validation, cleaning, revalidation, and run the analysis of the whole dataset. If desired, for a specific grouping, run *all_scenarios* and select how you want to group under *combo_vars. A*fter this, combine the results into one database and export, if needed, using the *write.csv* tool.

Here it selects where the functions are stored:

source("01_script/functions.R")

fish_data_raw <- read.csv("02_rawdata/generated_fish_data.csv", header = TRUE, stringsAsFactors = FALSE)

Here it select the validation function to run, and return its report:

Validate and return issues in the data, continue if no issues

validation <- validate_diet_data(fish_data_raw)

Validation

Here it select the validation tool, and saves a log of what was changed:

Cleaning step

clean_result <- clean_diet_data(fish_data_raw)

clean_data <- clean_result$data

clean_log <- clean_result$log

Here the validation tool is re-run to verify if the cleaning process was successful:

Validate using clean_data to show that the cleaning step worked

validate_diet_data(clean_data)

Here the data analysis takes place:

to run the whole pipe line

pipeline_result <- run_diet_pipeline(clean_data)

Here the user can select which scenario he wants to run, with the logic of maybe the scientist wanting to run a scenario where only selects a set of sampling sites/seasons/species:

Calculate the result of every possible scenario combination (Within your choice of, sampling_area, season, sex and species in "combo_vars")

all_scenarios <- run_all_observed_scenarios(

clean_data = clean_data,

combo_vars = c("sampling_area","season","sex","species"))

combined_scenarios <- dplyr::bind_rows(lapply(all_scenarios$results,function(x)x$combined),.id = "scenario_name")

Here the analysed data is exported in a .csv format.

Export the results in a .csv file

write.csv(combined_scenarios, "04_output/combined_scenarios.csv")

Here is a diagram of how the function *all_observed_scenarios* is structured:

```
Clean_data
        I
run_all_observed_scenarios()
        I
        + filtered scenario 1 - fo_summary()
        I                       - volume_summary()
        I                       - iai_summary()
        I                       - diet_indices_summary()
        I
        + filtered scenario 2 - fo_summary()
        I                       - volume_summary()
        I                       - iai_summary()
        I                       - diet_indices_summary()
        I
        + filtered scenario 3 - same analysis functions…
```

<u>Discussion</u>

It is difficult to produce a real-world comparison of an analysis made without the proposed package, but I foresee being able to help scientists spend less time writing and troubleshooting code and allowing them more time to spend interpreting their data. As a scientist myself, I understand the frustration of not being able to accomplish the results you expected. By simplifying the workflow and keeping it clear, trackable, and reproducible, it can help make diet data analysis more accessible to students and researchers.

What could be improved is to implement more index calculations, allowing scientists to use it not just to calculate the percentage of volume, occurrence, and importance alimentary index.

Estimations were made by fixing common mistakes, such as scientific name typos, misplaced commas, potential outlier errors and habitat that were input in error. It took around 25 minutes to fix it all manually. When using the automated tool, on the other hand, it took less than 5 seconds to run it all. About 300 times faster.

Conclusion

This work successfully outputs f FO%, V% and IAi% through a reproducible workflow that includes validation, cleaning, revalidation, and analysis of fish gut data. The automated approach reduces manual data-cleaning errors, preserves raw data history, and makes the analysis faster and more consistent.

In this project, tasks that took about 25 minutes manually were completed in under 5 seconds with the automated tool. More real-world testing is still needed, since this work used biased and randomly generated databases. Future versions could also include additional diet indices beyond FO%, V%, and IAi%.

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